

# HDB Technical Test for Senior Data Engineer

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|-----------------|-----------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>  | Technical test to assess engineering and solutioning competencies for a senior data engineer role in HDB. |
| <b>Duration</b> | <b>1 week</b> upon receiving the assignment                                                               |

## Assessment Focus

This assessment is not intended to reward the most elaborate or complex solution. Candidates should focus on demonstrating sound engineering judgement, clean and maintainable implementation, clear documentation of assumptions, and practical architecture thinking. Where trade-offs are involved, candidates should explain the rationale for their chosen approach.

## Part 1: Developing Data Pipelines

### Background

The Data Engineering Team is tasked to provide data cleaning and transformation services to the Data Science Team to aid their analysis of resale flat prices. Candidates need to design an ETL pipeline using datasets from January 2012 to December 2016.

All tasks should be performed programmatically where possible. Avoid manual interface interactions and hardcoding.

### Dataset Extraction

The required dataset can be found at <https://data.gov.sg/collections/189/view>. The ETL pipeline should process the data file as-is, without manual modifications such as manually deleting rows or columns or modifying the data values.

### Data Quality Requirements

1. Combine the datasets into a single master dataset. The combined dataset should contain all attributes found in all files.
2. Perform data profiling on the dataset. Candidates may code their own profiling rules or leverage open-source data profiling frameworks.
3. Design data validation rules to validate Date, Town, Flat Type, Flat Model, and storey\_range using Jan 2012 dataset as the authoritative set.
4. Assume HDB lease is 99 years old and compute the remaining lease accordingly, rounded down to Years and Months. [Do assumption where we take it as lease\\_commencement\\_date is 1st Jan of the year](#)
5. Assume the composite key for the dataset to be all the columns excluding resale price. Where duplicates are found, compare and take the higher resale price record as the reference.
6. Identify potentially anomalous resale price using appropriate heuristics.

7. Include any additional data profiling, cleaning or validation rules deemed appropriate. The actual dataset may not contain anomalous data, but candidates should demonstrate mechanisms to identify such behaviours.
8. Document insights and assumptions if any in a markdown file.

### Data Transformation Requirements

9. Using the cleaned data from the previous step, create a new column called Resale Identifier derived from the rules below.
  - a. Assume the composite key for the dataset to be all the columns excluding resale price. Where duplicates are found, compare and take the higher resale price record as the reference.
10. Hash the identifier column using an irreversible hashing algorithm while preserving uniqueness.
  - First character is S.
  - Next 3 digits to be the first 3 digits from the block column after removing any non-numerical characters. If the block column digits has less than 3 digits, to prepend the appropriate number of zeros. Example, block number 19 becomes 019.
  - Next 2 digits is from the 1st to 2nd digit of the average resale price based on the groupings of year-month, town and flat\_type. Example. if average Ang Mo Kio 2 room flat resale price is \$230000, the 2 digits will be 23.
  - The last two digits are the month of the current entry. For example, if the month is 2012-01, the last two digits are 01.
  - The final character is the first character of the town. Example: Ang Mo Kio becomes A.

### Data Output Requirements

11. Candidates may create any number of tables or views to aid the ETL process.
12. There are five mandatory output groups to be produced:

| Output group | Description                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw          | Contains the raw data files as-is.                                                                                                                |
| Cleaned      | Dataset that passes the data quality requirements.                                                                                                |
| Transformed  | Dataset that went through the transformation requirements.                                                                                        |
| Quarantined  | Dataset that contains records that do not meet the requirements such as records that fail validation checks, duplicate records or anomalous data. |
| Hashed       | Cleaned data with the hashed identifier column.                                                                                                   |

### Submission Requirements

13. ETL pipeline to be developed in Python. Candidates are free to design the pipelines in any manner deemed appropriate.

14. A Jupyter Notebook file with input files should be uploaded to a Git repository. The notebook should include in-line comments and markdown documentations to guide execution of the codes and production of the intended output. It should also contain cell outputs.
15. Ensure that the codes adhere to software engineering and data engineering best practices. Candidate should provide a README.md file at the root of the repository with sufficient information for the reviewers to understand, set up and execute without additional clarifications.

## Part 2: Architecting Data Ingestion and Data Exploitation Solution Patterns

### Background

After developing the pipeline, there is a need to operationalise it. This includes designing the architecture to support data ingestion and data exploitation, enabling an end-to-end flow of data to support analysis of the dataset.

Assume HDB's data platform is built on AWS with target data store as AWS S3. Design solution architectures that support the requirements. Use the AWS architecture icon pack to aid in architecture diagrams:

<https://aws.amazon.com/architecture/icons/>.

The design should take security, scalability, maintainability and performance into consideration. Candidates could make any necessary assumptions and document down the assumptions accordingly.

The objective of this part is to focus on architectural design and solutioning. Actual coding is not required.

### Data Ingestion Requirements

1. Batch data ingestion from data.gov.sg. The design should support large file transmission such as files > 100 MB.
2. data.gov.sg endpoint is public facing while HDB's data platform resides in private VPCs. The design should factor in network and system component segmentations as part of security consideration.

### Data Exploitation Requirements

The Data Science Team is using Tableau on AWS for analysis. It resides in another private VPC. The architecture should support:

3. Data integration via Athena Driver. Reference: [Tableau Amazon Athena connector documentation](#)
4. The integration design should factor in network security to keep network traffic flow private.

### Submission Requirements

5. Output could be done in power point, draw.io or any equivalent tools and converted to PNG for submission. Be as detailed as possible.
6. Together with Part 1, upload the architectural designs to the same Git repository and configure it to be publicly accessible.
7. Do not upload this test assignment into the Git repository.